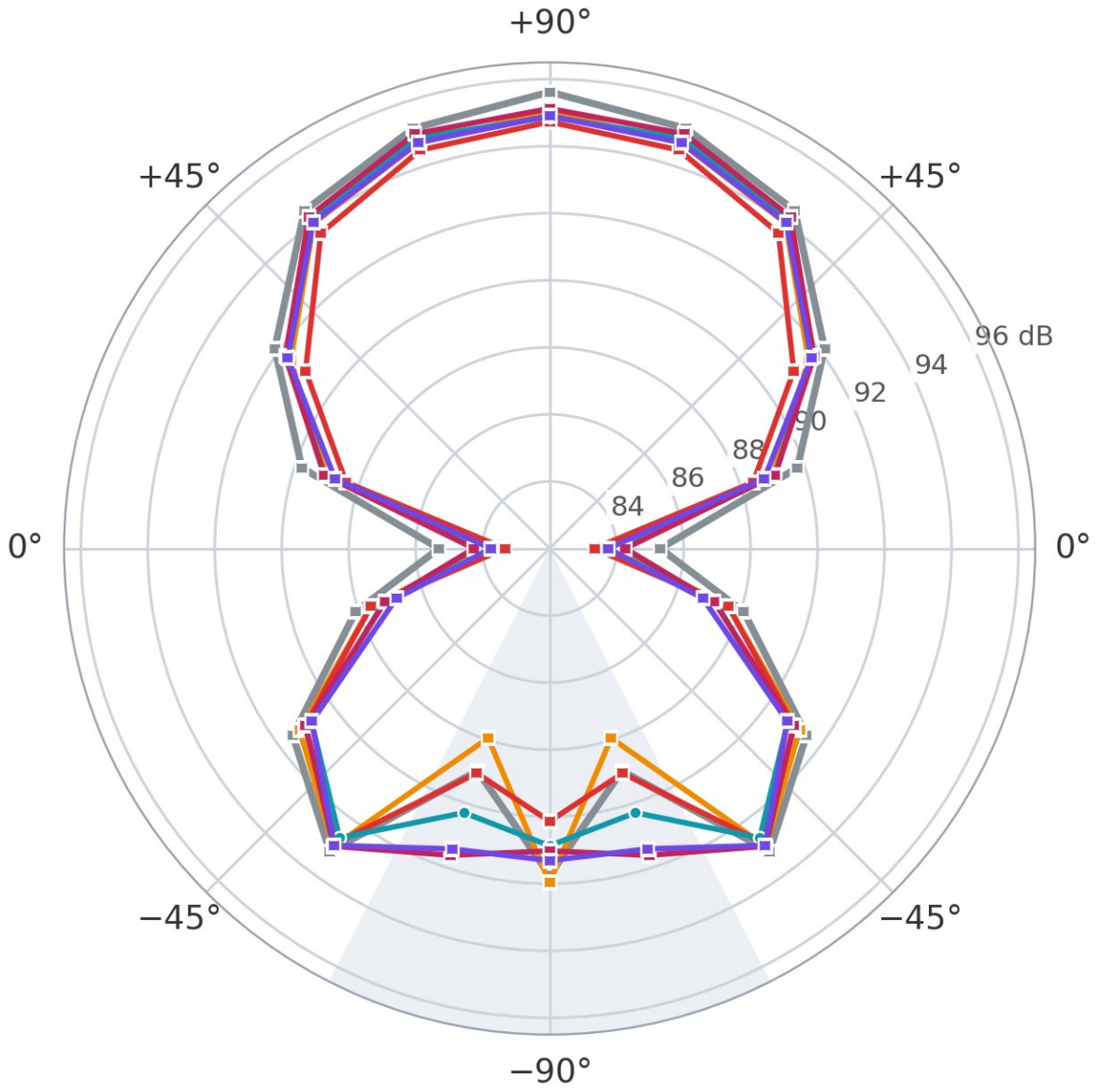


THE DUCT LININGS

Duct linings at 2000 µs

Mean level. Square markers: 2 Nov. Round: 30 Oct. Bare prop off the plot.

		level	thrust	rpm
Bare prop, no duct	2 Nov	80.0 dB	**10.56 N**	11,807
Thinsulate, 4 layers	2 Nov	90.6 dB	**9.22 N**	12,157
Thinsulate, 2 layers	2 Nov	90.9 dB	**8.78 N**	12,205
Thinsulate, 6 layers	30 Oct	91.0 dB	**8.64 N**	12,176
Thinsulate, 10 layers	2 Nov	91.0 dB	**8.89 N**	12,177
Thinsulate, 8 layers	2 Nov	91.2 dB	**8.82 N**	12,200
Duct, no lining	2 Nov	91.5 dB	**7.99 N**	12,269

+1.23 N

Thinsulate, 4 layers against the unlined duct: it wins back 48 % of the 2.57 N the empty duct loses to the bare prop.

Every duct is louder: 90.6–91.5 dB against 80.0 dB bare, and spins 12,157–12,269 rpm against 11,807 rpm bare (+3.0 to +3.9 %).

Thinsulate, 4 layers is also the quietest lining: 0.3–2.0 dB below the unlined duct from −54° to +90° (0.9 dB on average).

2 Nov 2026 (6 layers: 30 Oct), APC 8x4.5MR, AXI 2217/12, 500 Hz–24 kHz, 2000 µs, mean of 3 repeats. Axis 82–96.5 dB. Grey: −90°/−72°, probably in the flow. rpm = stand × 60/7.